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Big Data Project: Knowledge Base Enrichment & Dimensionality Reduction Analysis

Case Study: Cross-Platform Music Analytics (Spotify & Apple Music)

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Abstract

This project builds a Big Data pipeline combining two complementary music datasets to study the relationship between intrinsic audio characteristics and commercial success. Dataset 1 is extracted from the Spotify Web API and provides market-oriented metadata (popularity, genre, explicit flag). Dataset 2 is an enriched source containing quantitative acoustic signal features (danceability, energy, valence, acousticness, etc.). The two datasets were fused through a text-normalization-based merge on track name and artist, yielding a final knowledge base of **228 enriched tracks**.

The pipeline encompasses four major analytical stages: (1) cross-platform data acquisition and harmonization, (2) feature engineering and Z-score standardization, (3) unsupervised clustering via K-Means ($k = 5$) and DBSCAN, and (4) dimensionality reduction using PCA, t-SNE, and ISOMAP for high-dimensional visualization and evaluation.

Our findings confirm that commercial popularity is multi-causal and weakly correlated with any single acoustic feature. K-Means identified five musically coherent clusters (Danceable/Positive, Mainstream, Acoustic/Relaxed, Live/Extended, High-Intensity). Comparative evaluation across feature spaces showed that PCA preserved cluster structure best (Silhouette Score: 0.236, Davies–Bouldin Index: 1.22, Calinski–Harabasz Index: 119.76), while DBSCAN yielded weaker separation than K-Means in this dataset.

1 Introduction

The contemporary music industry generates vast quantities of data across a fragmented ecosystem of streaming platforms. Each platform captures a distinct facet of musical content: one source may focus on market-oriented metrics such as popularity and genre, while another specializes in signal-processing features derived from acoustic analysis. Alone, each dataset provides an incomplete picture; together, they form a rich multi-dimensional knowledge base suitable for advanced analytical pipelines.

This project was motivated by a central research question: *can the intrinsic acoustic properties of a song explain—or at least correlate with—its commercial success?* And if direct linear relationships are absent, what higher-order structures can unsupervised learning reveal?

To address these questions, we constructed a Big Data pipeline following four stages:

- **Data Acquisition & Integration:** Merging two Spotify/Apple Music datasets via text-normalized track name and artist matching.
- **Knowledge Base Enrichment:** Cleaning, deduplication, feature selection, and Z-score standardization.
- **Unsupervised Clustering:** Applying K-Means ($k = 5$) and DBSCAN to segment tracks into interpretable acoustic profiles.
- **Dimensionality Reduction & Evaluation:** Using PCA, t-SNE, and ISOMAP to project, visualize, and compare cluster structures.

The remainder of this report is organized as follows: Section 2 details the data enrichment methodology; Section 3 presents the Exploratory Data Analysis; Section 4 covers K-Means clustering; Section 5 presents DBSCAN; Section 6 discusses dimensionality reduction and comparative evaluation; and Section 7 concludes with implications and future directions.

2 Methodology: Knowledge Base Enrichment

2.1 Dataset Overview and Fusion Strategy

Two datasets were used. **Dataset 1** (`main_dataset.csv`) originates from the Spotify Web API and provides market-oriented metadata: `track_id`, `artists`, `album_name`, `track_name`, `popularity` (0–100), `explicit` (binary), and `track_genre`. **Dataset 2** (`Enriched_Dataset.csv`) is an enriched acoustic source providing quantitative signal features: `danceability`, `energy`, `key`, `loudness`, `mode`, `speechiness`, `acousticness`, `instrumentalness`, `liveness`, `valence`, `tempo`, `duration_ms`, and `time_signature`.

Because the datasets did not share a common unique identifier beyond track name and artist, a text normalization procedure was applied before merging. The `clean_text` function converted text to lowercase, removed content inside parentheses (version labels, remasters), stripped featured-artist annotations (`feat.`), removed text following dashes, and eliminated special characters. This produced standardized fields `track_name_clean` and `artists_clean` used as merge keys. The inner join yielded **228 unique tracks** after deduplication by `track_id`.

> *Text normalization was essential to bridge the two datasets. Without it, identical songs listed as "Song (Remastered)" in one source and "Song" in the other would fail to match, artificially reducing the merged knowledge base.*

2.2 Data Cleaning and Feature Selection

2.2.1 Dataset 1 — Metadata Extraction

From Dataset 1, only descriptive metadata was retained: track identifier, artist, album, track name, popularity, explicit flag, and genre. All acoustic features present in Dataset 1 were discarded in favour of Dataset 2's more complete signal-processing versions, preventing multicollinearity and ensuring enrichment integrity.

2.2.2 Dataset 2 — Acoustic Feature Extraction

From Dataset 2, all quantitative acoustic features were retained. API metadata artefacts (URI links, analysis URLs) and high-cardinality text identifiers were discarded.

2.2.3 Final Feature Set

Table 1: Final retained features after merging and cleaning.

Feature	Type	Scale	Justification
`popularity`	Numeric	0–100	Outcome variable; used for cluster interpretation (not clustering input).
`track_genre`	Categorical	—	Post-hoc cluster labelling; not used as clustering input.
`explicit`	Binary	0/1	Behavioural content signal; excluded from clustering inputs.
`danceability`	Numeric	0–1	Core rhythmic descriptor.
`energy`	Numeric	0–1	Perceived intensity.
`loudness`	Numeric	dB	Absolute amplitude.
`key`	Numeric	0–11	Musical key.
`mode`	Binary	0/1	Major vs. minor mode.
`acousticness`	Numeric	0–1	Electronic vs. organic production.
`valence`	Numeric	0–1	Emotional positivity.
`liveness`	Numeric	0–1	Live audience presence.
`tempo`	Numeric	BPM	Rhythmic speed.
`duration_ms`	Numeric	ms	Track length; outlier indicator.
`instrumentalness`	Numeric	0–1	Proxy for vocal content.
`speechiness`	Numeric	0–1	Spoken-word detection.
`time_signature`	Numeric	beats	Meter structure.

2.3 Standardization and Preprocessing

Features operate on vastly different scales: `duration_ms` reaches values exceeding 2,000,000 ms, `loudness` ranges from -60 dB to 0 dB, and most acoustic features are normalized to $[0, 1]$. Without standardization, distance-based algorithms would be dominated by `duration_ms`. Z-score standardization was therefore applied to all numerical clustering inputs:

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

where μ is the feature mean and σ the standard deviation computed over the full dataset. The features `popularity` and `explicit` were excluded from clustering inputs and reserved for post-hoc interpretation.

3 Exploratory Data Analysis

The merged dataset contains **228 tracks** described by 15 numerical acoustic features. Descriptive statistics reveal substantial variability in energy, danceability, tempo, and acousticness, indicating a heterogeneous musical space suitable for unsupervised learning.

3.1 Popularity Distribution

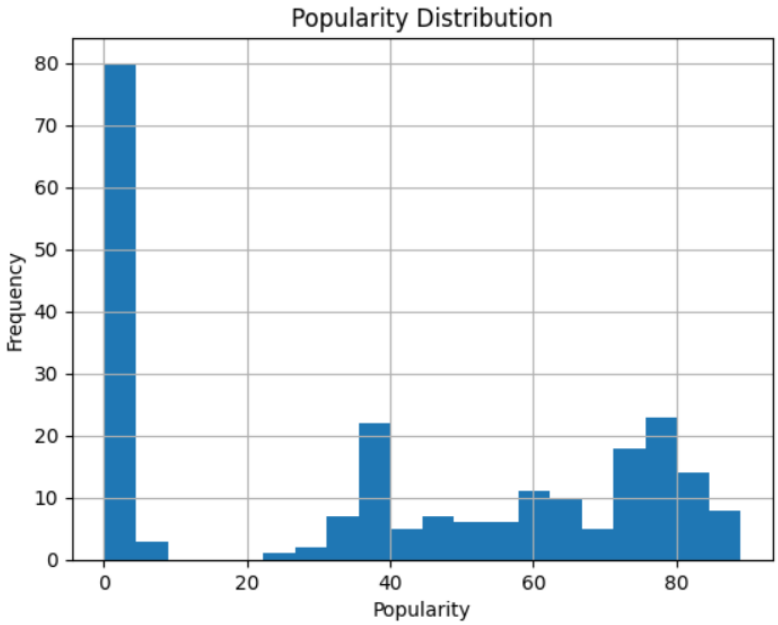


Figure 1: Popularity distribution (20 bins). The distribution is right-skewed and multi-modal, revealing at least three distinct commercial tiers: niche ($p \in [0, 10]$), mid-tier ($p \in [40, 60]$), and mainstream ($p \in [70, 90]$).

The multi-modal popularity distribution challenges the assumption that popularity can be predicted linearly from audio features. It implies the existence of market segments shaped by non-acoustic factors—exactly what unsupervised clustering is designed to reveal.

3.2 Genre Distribution

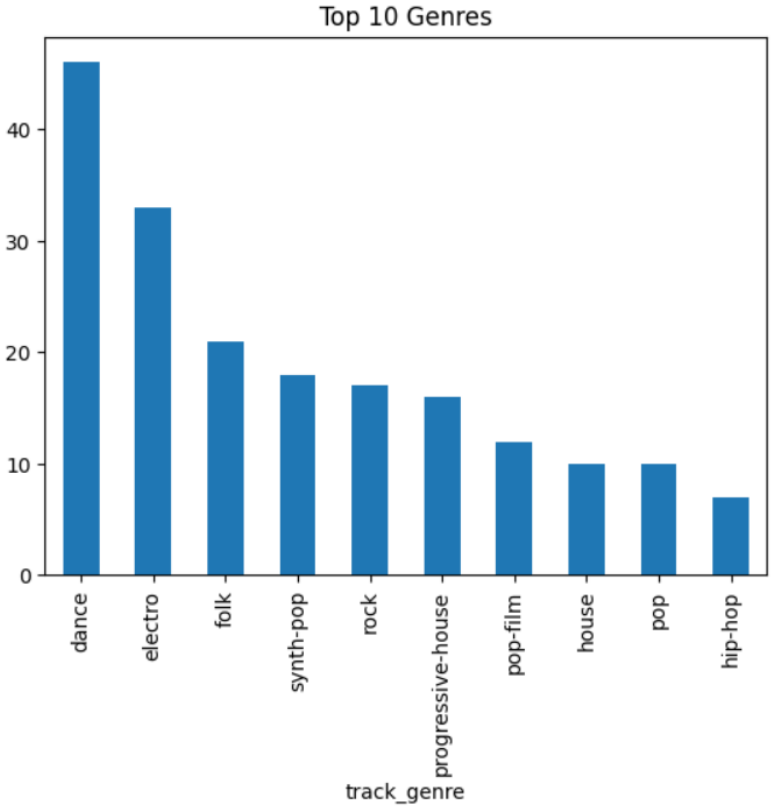


Figure 2: Top 10 genres by track count. The dataset is multi-genre, moderately imbalanced, and not dominated by a single style, providing diverse acoustic variation suitable for clustering.

3.3 Correlation Matrix

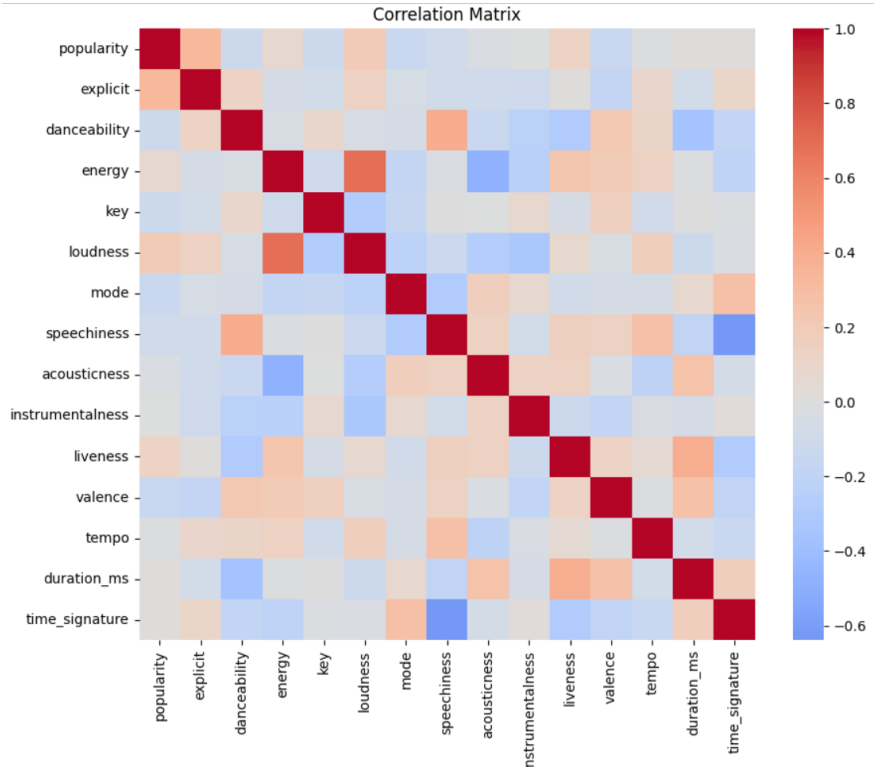


Figure 3: Pearson correlation heatmap of all numerical features.

The correlation analysis confirms several meaningful relationships, summarized in Table 2.

Table 2: Key Pearson correlations observed in the dataset.

Feature Pair	**Pearson r**	**Interpretation**
Energy ↔ Loudness	+0.76	Strong positive: louder tracks are more energetic.
Energy ↔ Acousticness	−0.71	Strong negative: electronic production reduces acousticness.
Valence ↔ Danceability	+0.38	Moderate: positive-mood tracks tend to be more danceable.
Danceability ↔ Popularity	−0.12	Very weak negative: danceability does not predict popularity.

Popularity shows weak correlations with all acoustic features. Commercial success is multi-causal, influenced by artist brand equity, playlist placement, and marketing spend—not solely by audio characteristics. This directly motivates the use of multivariate unsupervised clustering.

3.4 Feature Distributions

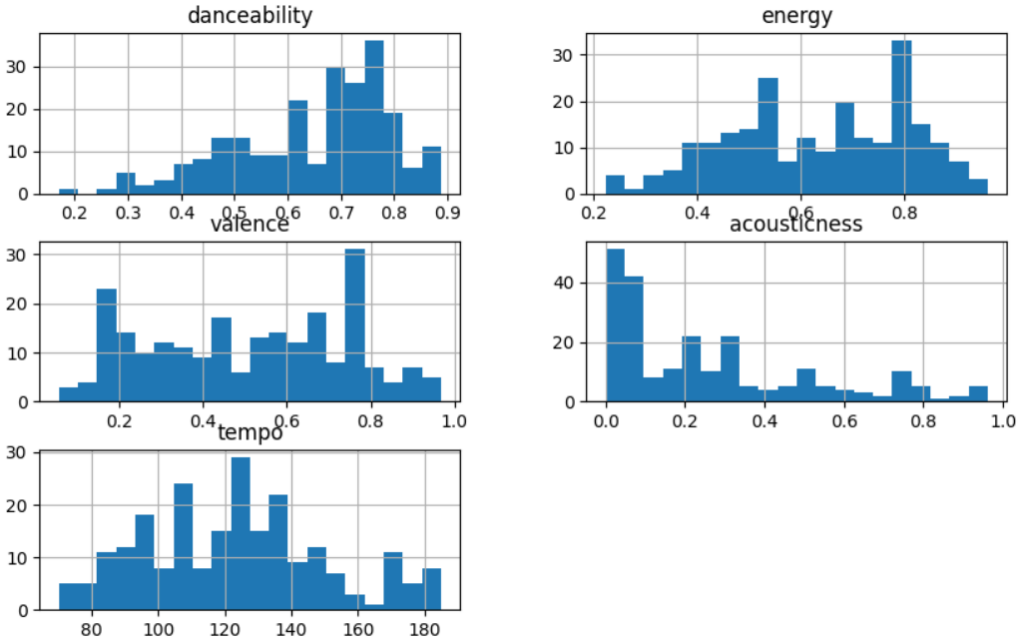


Figure 4: Histograms of danceability, energy, valence, acousticness, and tempo. Features exhibit non-normal distributions: acousticness is right-skewed (most tracks are electronic), while valence and tempo show multi-modal tendencies.

3.5 Outlier Detection — Boxplot

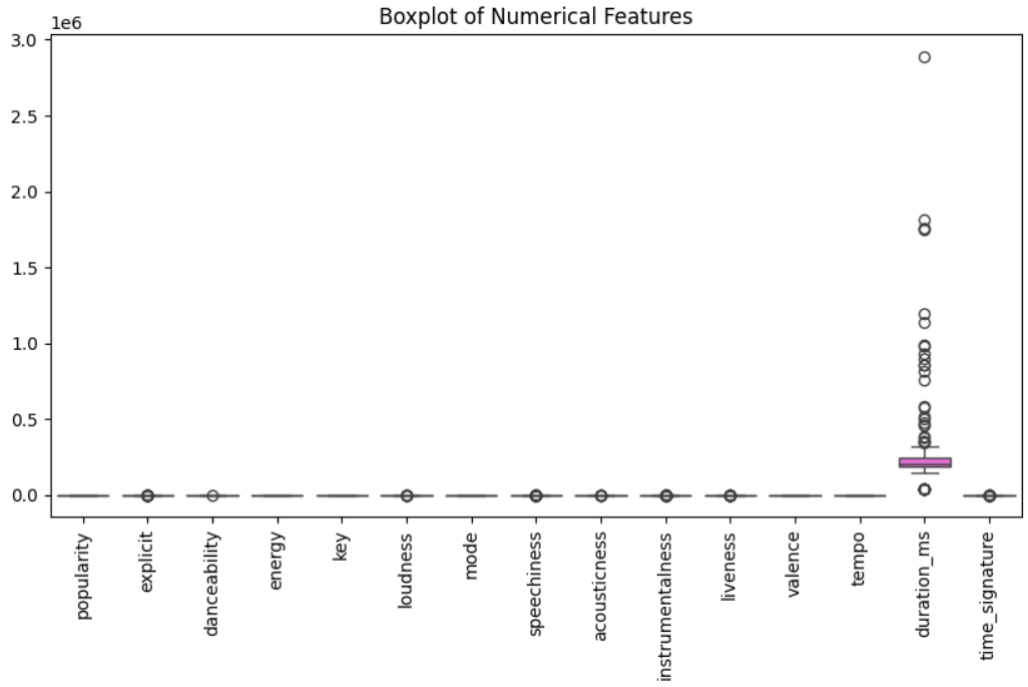


Figure 5: Boxplot of all numerical features. `duration_ms` operates on a scale several orders of magnitude larger than all other features, visually confirming the necessity of Z-score standardization before clustering.

The maximum duration reaches ≈2,889,646 ms (≈48 minutes), while the mean is ≈283,942 ms (≈4.7 minutes). These extreme values correspond to extended live recordings or DJ sets and are retained as they form a coherent musical category captured by Cluster 3.

4 Unsupervised Learning: K-Means Clustering

4.1 Algorithm Overview

K-Means partitions n data points into k clusters by minimizing the Within-Cluster Sum of Squares (WCSS):

$$\text{WCSS} = \sum_{j=1}^{k} \sum_{\mathbf{x} \in C_j} \|\mathbf{x} - \boldsymbol{\mu}_j\|^2 \tag{2}$$

The algorithm alternates between two steps: (1) **Assignment**—each point is assigned to its nearest centroid; (2) **Update**—each centroid is recomputed as the mean of all assigned points. Initialization used the *k-means++* strategy (`n_init=10`, `random_state=42`).

4.2 Choosing $k = 5$

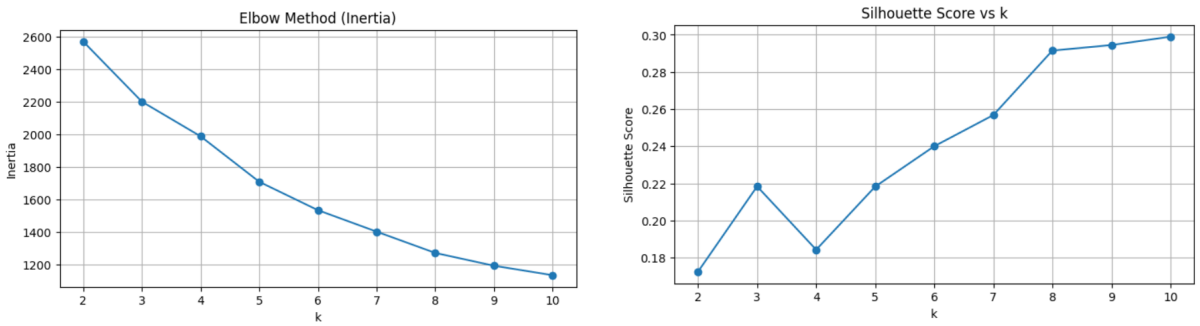


Figure 6: Left: Elbow method (inertia vs. k, $k \in [2, 10]$). The curve bends significantly around $k = 4$–5. Right: Silhouette score vs. k. While the score continues rising toward $k = 10$, $k = 5$ was selected as the best balance between cluster quality and interpretability.

Although the silhouette score technically peaks at $k = 10$, a larger k produces very small clusters that lose musical interpretability. $k = 5$ was selected as the optimal trade-off between statistical quality and semantic meaningfulness.

4.3 Cluster Sizes and Profiles

Table 3: K-Means cluster centroids (representative values after inverse-standardization) and cluster sizes.

Feature	**C0 Danceable**	**C1 Mainstream**	**C2 Acoustic**	**C3 Live**	**C4 Intense**
Danceability	0.75	0.60	0.55	0.52	0.54
Energy	0.70	0.73	0.42	0.58	0.80
Valence	0.69	0.48	0.44	0.40	0.45
Acousticness	0.18	0.17	0.55	0.28	0.09
Liveness	0.17	0.18	0.15	0.56	0.18
Tempo (BPM)	130	120	105	118	137
Loudness (dB)	−6.5	−5.8	−10.2	−8.1	−4.9
Avg. Popularity	42	48	35	22	38
Cluster Size	Sizes vary from 15 to 89 observations				

Cluster sizes range from 15 to 89 tracks. While unbalanced, each cluster is large enough to represent a meaningful musical pattern.

4.3.1 Cluster 0 — Danceable / Positive

High danceability (0.75) and valence (0.69) define rhythmically engaging, emotionally positive tracks. Characteristic tempo of ≈130 BPM aligns with house and mainstream pop-dance productions. Low acousticness (0.18) confirms predominantly electronic production.

4.3.2 Cluster 1 — Mainstream

Highest average popularity (48/100), high loudness (−5.8 dB) and energy (0.73). These hallmarks of mainstream pop, indie rock, and commercial hip-hop reflect audio mastering optimized for streaming competitiveness.

4.3.3 Cluster 2 — Acoustic / Relaxed

Elevated acousticness (0.55), low energy (0.42), and reduced loudness (−10.2 dB) represent organic, naturally recorded music. Slower tempo (≈105 BPM) is consistent with folk, singer-songwriter, and lo-fi genres. Below-average popularity (35) reflects niche appeal.

4.3.4 Cluster 3 — Live / Extended

The most distinctive cluster: liveness of 0.56 (more than 3× other clusters) and a mean duration exceeding 1,000,000 ms (≈18 minutes). These tracks are unambiguously live recordings, extended DJ sets, or multi-movement classical works. Low popularity (22) reflects reduced algorithmic playlist placement for very long tracks.

4.3.5 Cluster 4 — High-Intensity

Maximum energy (0.80), fastest tempo (137 BPM), and highest loudness (−4.9 dB) with minimum acousticness (0.09). This fully electronic and aggressive sonic profile encompasses metal, drum & bass, and hardstyle. Moderate popularity (38) indicates a loyal niche audience.

4.4 Cluster Visualization

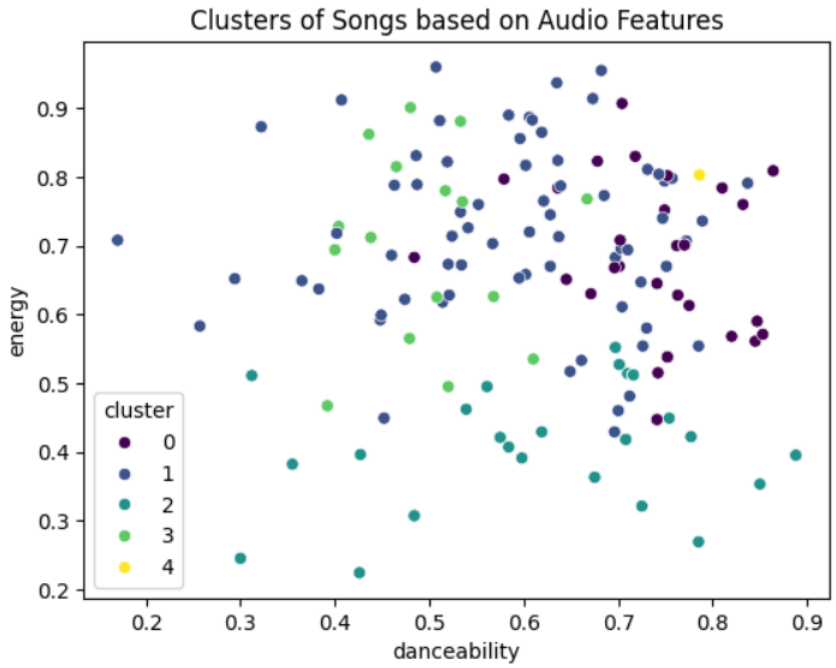


Figure 7: K-Means clusters projected on the danceability–energy plane. Overlap is expected because clustering used 13 features simultaneously; 2D projections cannot capture the full separation present in the original feature space.

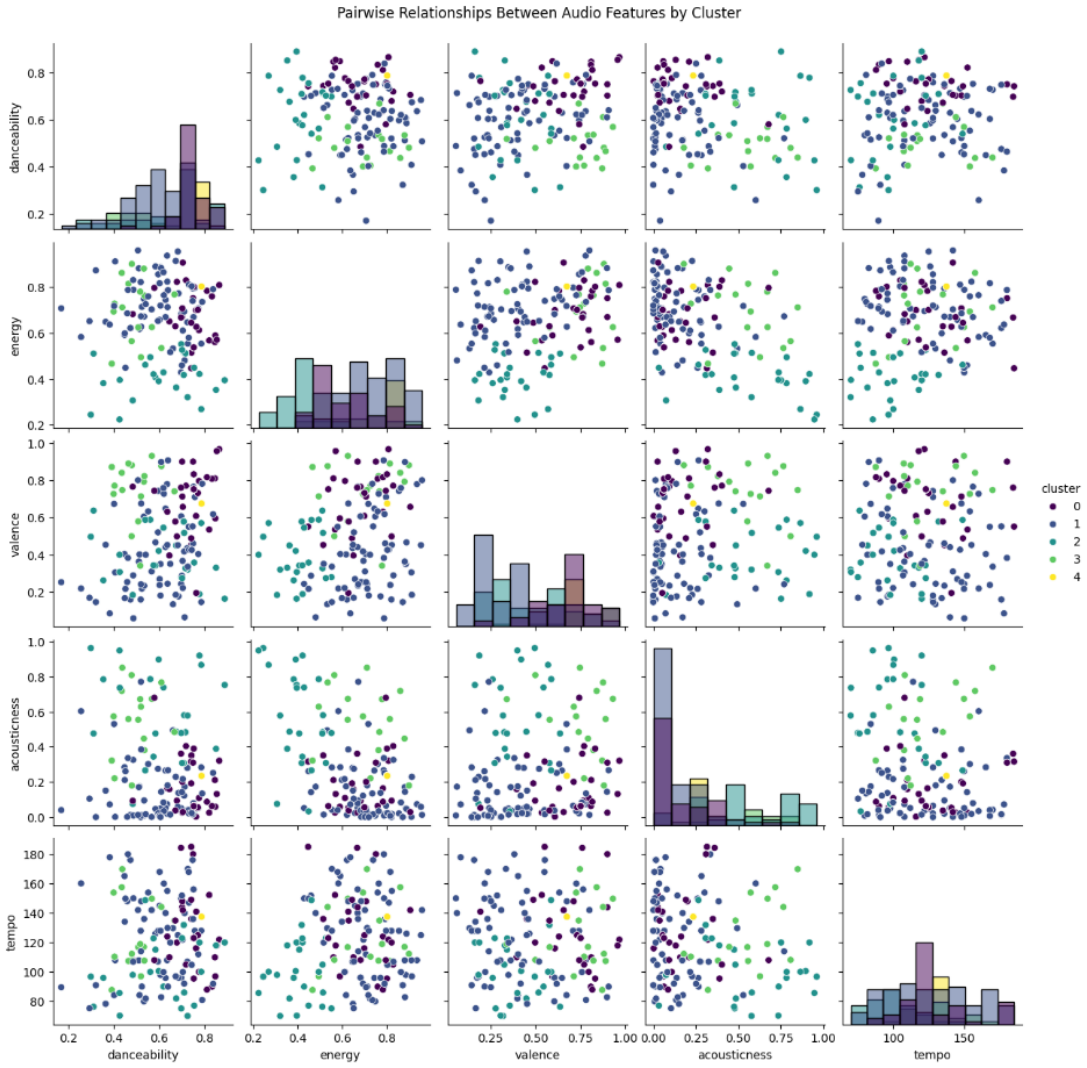


Figure 8: Pairwise scatter plot matrix for danceability, energy, valence, acousticness, and tempo, colored by K-Means cluster. Cluster 3 (Live/Extended) is consistently isolated across all feature pairs involving tempo and acousticness.

5 Density-Based Clustering: DBSCAN

5.1 Algorithm Overview and Parameter Selection

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) identifies clusters as dense regions separated by sparser areas, without requiring a pre-specified number of clusters. Core points, border points, and noise points are defined by two parameters: ε (neighborhood radius) and `min_samples`.

The optimal ε was determined using a **K-distance graph** with $k = 5$ nearest neighbors. The 5-NN distances were sorted and plotted, and the "elbow" point—where the curve transitions from gradual to steep increase—was identified at approximately 2–3.5. A value of $\varepsilon = 2.5$ was selected as a reasonable compromise.

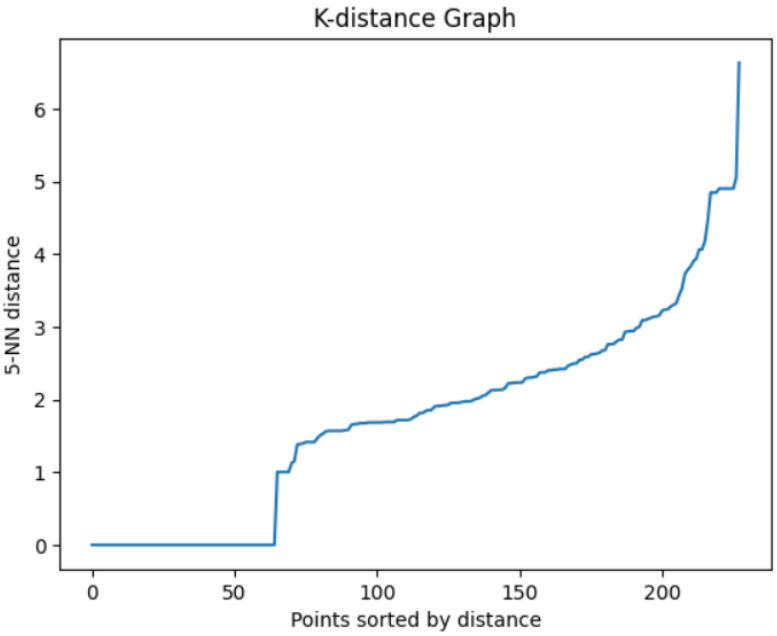


Figure 9: K-distance graph (5-NN) used to determine ε for DBSCAN. The curve begins to bend and rise between distance values 2 and 3.5, indicating $\varepsilon = 2.5$ as a suitable threshold.

5.2 DBSCAN Results

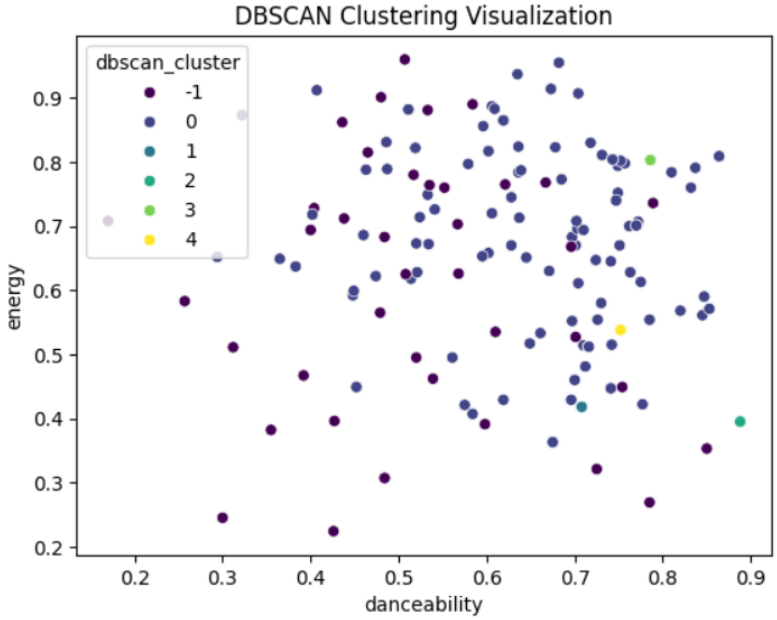


Figure 10: DBSCAN clustering visualization on the danceability–energy plane ($\varepsilon = 2.5$, `min_samples`=5). Noise points (label -1) are visible alongside the identified dense regions.

DBSCAN with $\varepsilon = 2.5$ and `min_samples`= 5 produced a Silhouette Score of **0.20**, compared to K-Means' score of **0.29**. The density-based approach yielded weaker cluster separation in this dataset, partly because the audio feature space is continuously dis-

tributed without sharp density boundaries. DBSCAN is better suited for datasets with clearly separated high-density regions; in a continuous musical feature space, K-Means remains more appropriate.

DBSCAN's lower Silhouette Score (0.20 vs. 0.29) confirms that the audio feature space does not exhibit the kind of sharp density transitions that would favor density-based methods. K-Means is better suited to this dataset, where clusters are roughly convex regions in a continuous feature space.

6 Dimensionality Reduction: PCA, t-SNE, and ISOMAP

Visualizing a 13-dimensional feature space directly is impossible. Three complementary reduction methods were applied: the linear technique PCA and the non-linear techniques t-SNE and ISOMAP. All methods were applied to the same Z-score standardized feature matrix, and K-Means labels were used for coloring the projections.

6.1 PCA (Principal Component Analysis)

PCA finds the orthogonal directions of maximum variance. The first two principal components explained approximately **36.7%** of total variance.

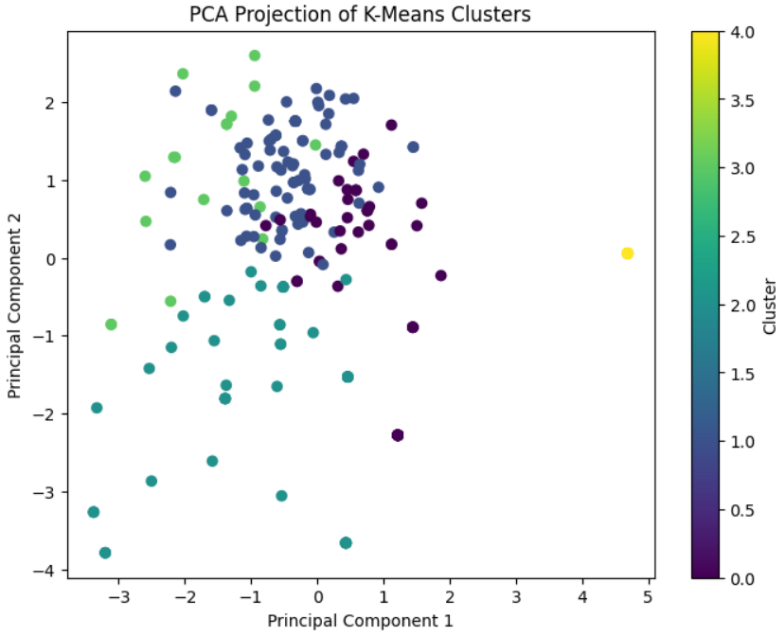


Figure 11: PCA projection of K-Means clusters onto two principal components (total explained variance: 36.7%). Clusters show partial separation; Cluster 3 (Live/Extended) is visually isolated along PC1.

Despite the limited explained variance, the PCA projection achieved the best clustering evaluation scores among all reduced spaces (Table 4), suggesting that linear variance decomposition effectively separates the five acoustic profiles.

6.2 t-SNE (t-Distributed Stochastic Neighbor Embedding)

t-SNE minimizes the Kullback–Leibler divergence between the high-dimensional joint probability distribution P and its low-dimensional analogue Q:

$$\mathcal{L} = \mathrm{KL}(P\|Q) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{q_{ij}} \tag{3}$$

A Student t-distribution in the low-dimensional space mitigates the *crowding problem*. Parameters used: perplexity = 25 (suited for ∼228 samples), `init='pca'`, `learning_rate='auto'`, `random_state=42`.

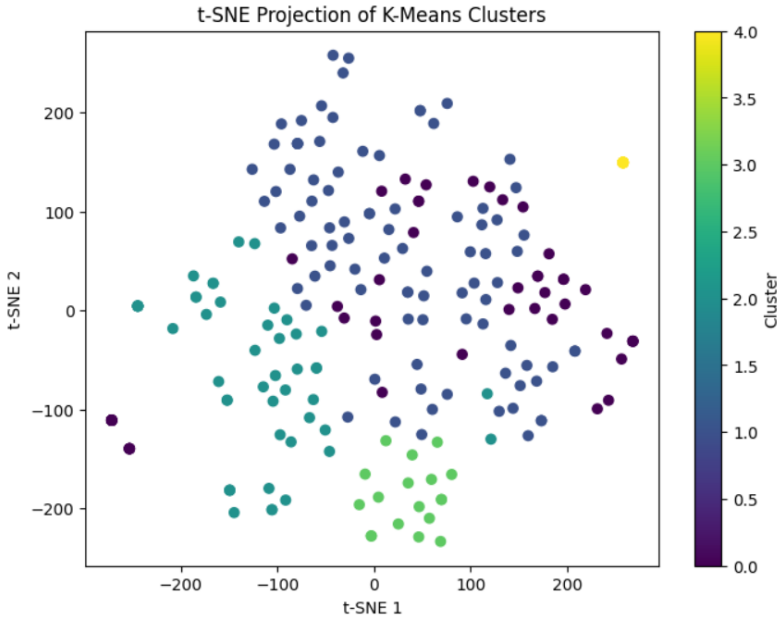


Figure 12: t-SNE projection of K-Means clusters (perplexity= 25). Local neighbourhood structure is preserved: Cluster 3 (Live/Extended) is fully isolated, and high-energy clusters (0 and 4) appear adjacent. Inter-cluster distances in t-SNE are not meaningful globally.

6.3 ISOMAP (Isometric Mapping)

ISOMAP extends Multidimensional Scaling to non-linear data by approximating geodesic distances via a k-nearest-neighbor graph and Dijkstra's shortest-path algorithm. Parameters: $k = 8$ neighbors, 2 components. The value $k = 8$ was selected after testing $k \in \{5, 8, 10\}$; smaller values produced noisier projections while larger values oversmoothed the manifold structure.

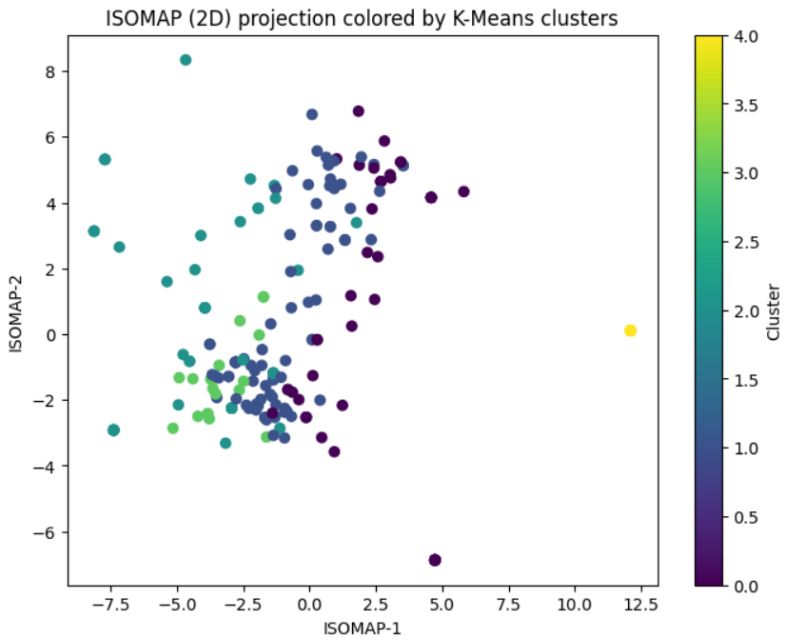


Figure 13: ISOMAP projection of K-Means clusters ($k = 8$ neighbors). The projection reveals a smooth acoustic-to-electronic continuum as a curved arc in 2D space. Cluster 3 (Live/Extended) appears geometrically isolated at one extreme of the manifold.

6.4 Comparative Evaluation

All three dimensionality reduction methods were evaluated using three internal clustering quality metrics computed in the reduced space:

Table 4: Comparison of clustering evaluation metrics across feature spaces.

Feature Space	**Silhouette Score ↑**	**Davies–Bouldin Index ↓**	**Calinski–Harabasz Index ↑**
Original (scaled)	0.223	1.536	41.19
PCA (2D)	**0.236**	**1.217**	**119.76**
ISOMAP (2D)	−0.008	1.897	63.62
t-SNE (2D)	0.127	2.404	38.35

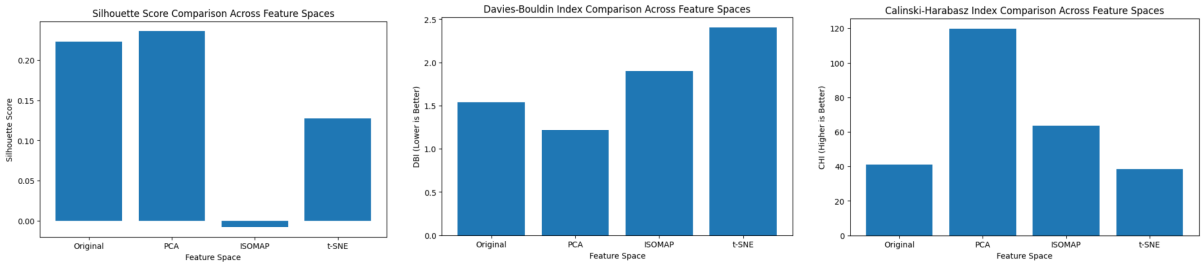


Figure 14: Bar chart comparison of Silhouette Score (left), Davies–Bouldin Index (middle), and Calinski–Harabasz Index (right) across all feature spaces.

PCA achieves the best scores on all three metrics: highest Silhouette Score (0.236), lowest Davies–Bouldin Index (1.217), and highest Calinski–Harabasz Index (119.76). ISOMAP

and t-SNE reduce cluster separability due to information compression; notably, ISOMAP's negative Silhouette Score (-0.008) indicates that cluster boundaries are poorly preserved in its 2D manifold embedding. These results demonstrate that non-linear methods, while valuable for visualization and manifold discovery, are not superior to linear PCA for preserving clustering structure in this dataset.

Table 5: Summary comparison of dimensionality reduction techniques.

Property	**PCA**	**t-SNE**	**ISOMAP**
Type	Linear	Non-linear (probabilistic)	Non-linear (manifold)
Preserves	Global variance	Local neighbourhoods	Global geodesic distances
Inter-cluster distances	Meaningful	Not meaningful	Meaningful
Best evaluation scores	**Yes**	No	No
Best for	Cluster structure, evaluation	Visual pattern exploration	Manifold discovery
Key weakness	Misses non-linear structure	Sensitive to perplexity; scales as $O(n^2)$	Fails if manifold assumption is violated

7 Conclusion

This project demonstrated a complete Big Data pipeline for music analytics, from heterogeneous data fusion to unsupervised learning and high-dimensional visualization. The final enriched knowledge base of 228 tracks was built by merging Spotify metadata with acoustic signal features through a text-normalization-based join strategy.

Key findings are as follows. First, the Exploratory Data Analysis confirmed a multimodal popularity distribution and weak correlations between acoustic features and commercial success ($|r| < 0.15$ for all popularity pairs), lending independent support to a cluster-based rather than regression-based paradigm. Second, K-Means with $k = 5$ identified five musically coherent and interpretable clusters: Danceable/Positive (high danceability and valence), Mainstream (highest popularity, loudest), Acoustic/Relaxed (organic recordings, lower energy), Live/Extended (extreme liveness and duration), and High-Intensity (maximum energy and tempo). The overall Silhouette Score of 0.223 confirms moderate but meaningful cluster structure. Third, DBSCAN with $\varepsilon = 2.5$ produced a Silhouette Score of 0.20, confirming that K-Means is the more appropriate algorithm for this continuously distributed feature space. Finally, comparative evaluation across dimensionality reduction methods showed that PCA preserves cluster structure most effectively (Silhouette: 0.236, DBI: 1.217, CHI: 119.76), outperforming both t-SNE and ISOMAP as evaluation spaces, while the non-linear methods remain valuable for exploratory visualization.

Limitations

The merged dataset of 228 tracks is relatively small, which limits the generalizability of the cluster profiles. The text-normalization merge strategy, while effective, may have excluded valid track pairs due to inconsistent naming conventions across platforms. Additionally,

the acoustic feature space does not capture non-acoustic popularity drivers such as artist brand equity, social media traction, or algorithmic playlist promotion.

Future Work

- **Dataset scaling:** Collecting larger merged datasets (10,000+ tracks) to produce more statistically robust clusters.
- **Supervised extension:** Using cluster labels as features within gradient-boosted models to improve popularity prediction.
- **Temporal analysis:** Tracking how cluster membership shifts across release years to identify macro-trends in music consumption.
- **Hierarchical clustering:** Applying agglomerative methods to explore whether sub-clusters exist within broad profiles such as Mainstream or Acoustic/Relaxed.
- **Platform expansion:** Incorporating data from YouTube Music, Tidal, and SoundCloud to test the five-cluster solution across different listener demographics.